

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Data Interpretation

Course Code: DSA0405

Course Title: FUNDAMENTALS OF DATA SCIENCE

CO Assessed: CO3 – Data Interpretation

Assessment: Assessment Tool 2 – Data Interpretation

Weightage: 40% of CIA

Total Marks: 100

CO3 Statement: Analyse and extract image features using detection and matching techniques for effective image representation and comparison. (BL4)

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Section / Batch: AI DATA SCIENCE

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Code of Conduct

I **B Tharun Kumar** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: B Tharun Kumar

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

Descriptive & Inferential Statistics – Assignment Answers

1. Sampling Distribution Reasoning

The Setup

We are told the population from which we are sampling has mean $\mu = 500$ and standard deviation $\sigma = 80$, and that a random sample of size $n = 64$ is drawn from it. The question asks us to reason about the behaviour of the sample mean $\bar{X}$ as a random variable in its own right — not about any single observation, but about how $\bar{X}$ itself would vary if we repeated the sampling process over and over.

(a) Standard Error of the Sample Mean

The standard error (SE) measures how much the sample mean is expected to fluctuate from sample to sample around the true population mean. It is not the same as the population standard deviation σ , which measures spread among individual observations — SE measures spread among sample means, and it shrinks as the sample size grows because averaging over more observations cancels out more of the individual randomness.

$$SE = \sigma / \sqrt{n}$$

$$SE = 80 / \sqrt{64} = 80 / 8 = 10$$

So the standard error of the sample mean is 10. This tells us that if we repeatedly drew samples of size 64 from this population and computed the mean each time, those sample means would themselves form a distribution centred at 500 with a spread (standard deviation) of about 10 — considerably tighter than the spread of 80 seen among individual observations in the population.

(b) Probability that the Sample Mean Exceeds 520

By the Central Limit Theorem, even if the underlying population distribution is not itself perfectly normal, the sampling distribution of $\bar{X}$ becomes approximately normal once n is reasonably large ($n = 64$ comfortably satisfies the usual rule of thumb of $n \geq 30$). So we can treat $\bar{X}$ as approximately $\text{Normal}(\mu = 500, SE = 10)$ and convert the raw value 520 into a Z-score to use the standard normal table.

$$Z = (\bar{X} - \mu) / SE$$

$$Z = (520 - 500) / 10 = 20 / 10 = 2$$

A sample mean of 520 is therefore exactly 2 standard errors above the population mean of 500. From the standard normal table, the area to the left of $Z = 2$ is 0.9772, so the area to the right (i.e., the probability of exceeding it) is:

$$P(\bar{X} > 520) = 1 - P(Z \leq 2) = 1 - 0.9772 = 0.0228$$

So there is approximately a 2.28% probability that a sample of 64 observations would produce a mean exceeding 520.

Practically, this is a fairly low probability — it tells us that observing a sample mean as high as 520 would be a somewhat unusual (though not extraordinarily rare) event if the population mean truly is 500. In hypothesis-testing terms, this is the kind of reasoning that underlies a one-sided test: if we actually observed $\bar{X} = 520$ in real data, a 2.28% chance of this happening purely by sampling variation would often be considered small enough to raise suspicion that the true population mean might actually be higher than 500.

(c) Effect of Increasing the Sample Size to $n = 256$

$$\text{New SE} = \sigma / \sqrt{n} = 80 / \sqrt{256} = 80 / 16 = 5$$

Increasing the sample size from 64 to 256 — a fourfold increase — causes the standard error to fall from 10 to 5, i.e., it is exactly halved. This is not a coincidence; it is a direct and general consequence of the formula $SE = \sigma / \sqrt{n}$.

Principle — the Inverse Square-Root Law: Because the sample size n sits under a square root in the denominator, the standard error does not shrink in direct proportion to n ; it shrinks in proportion to $\sqrt{n}$. To cut the standard error in half, the sample size must be increased by a factor of four (since $\sqrt{4} = 2$); to cut it to a third, n must increase ninefold; and so on. This is sometimes called the "law of diminishing returns" in sampling: the first handful of additional observations buys a great deal of extra precision, but as the sample grows large, each additional observation contributes progressively less reduction in SE. This principle is a direct consequence of the Central Limit Theorem — averaging over more independent observations causes individual random fluctuations to cancel out more thoroughly, so the sample mean settles more tightly around the true population mean — but it also explains why, in real research, doubling a study's precision is far more expensive (in terms of required sample size) than it might naively seem.

2. Covariance Calculation and Sign Interpretation

The Data

X (Advertising spend, ₹'000): 10, 20, 30, 40, 50 Y (Sales, ₹ lakh): 15, 18, 25, 30, 42

Covariance measures whether two variables tend to move together (both rise and fall in tandem), move in opposite directions, or show no consistent joint pattern at all. It is calculated as the average of the products of each variable's deviation from its own mean.

Step 1: Compute the Means

$$\bar{X} = \Sigma X / n \quad \bar{Y} = \Sigma Y / n$$

$$\bar{X} = (10+20+30+40+50)/5 = 150/5 = 30$$

$$\bar{Y} = (15+18+25+30+42)/5 = 130/5 = 26$$

Step 2: Deviations from the Mean

$$(X - \bar{X}): -20, -10, 0, 10, 20$$

$$(Y - \bar{Y}): -11, -8, -1, 4, 16$$

Notice that as X's deviation moves from negative to positive (as advertising spend rises through its range), Y's deviation follows the same pattern, moving from negative to positive as well — an early visual hint that the relationship will turn out to be positive.

Step 3: Products of Deviations and Their Sum

$$(-20)(-11) = 220, \quad (-10)(-8) = 80, \quad (0)(-1) = 0, \quad (10)(4) = 40, \quad (20)(16) = 320$$

$$\Sigma(X-\bar{X})(Y-\bar{Y}) = 220 + 80 + 0 + 40 + 320 = 660$$

Every product here is either positive or zero — there is not a single negative term — which is a strong sign that X and Y are consistently moving together across the entire dataset, not just on average.

Step 4: Covariance

$$\text{Cov}(X,Y) = \Sigma(X-\bar{X})(Y-\bar{Y}) / n$$

$$\text{Cov}(X,Y) = 660 / 5 = 132$$

Cov(X, Y) = 132.

Interpretation of the Sign

The covariance is positive, which means X (advertising spend) and Y (sales) tend to move in the same direction — as advertising spend increases, sales also tend to increase, and as advertising spend is low, sales tend to be low as well. This direction is entirely consistent with business intuition: advertising is expected to stimulate demand, so a positive relationship between spend and sales is exactly what a marketer would hope and expect to see. Had the covariance instead come out negative, it would have signalled the counter-intuitive and troubling possibility that higher spending was associated with lower sales, which would prompt investigation into whether the advertising campaigns were poorly targeted or otherwise ineffective.

What the Magnitude Alone Fails to Tell Us

While the sign of covariance is meaningful, its magnitude is not directly interpretable in isolation. Covariance is expressed in the product of the two variables' original units — here, ₹'000 (thousands) multiplied by ₹ lakh — so the number 132 is tied to this specific, somewhat arbitrary unit combination. If advertising spend had instead been recorded in ₹ (rupees) rather than ₹'000, the covariance would balloon by a factor of 1,000 even though the

underlying relationship between the variables has not changed at all. This means a covariance of 132 cannot, by itself, tell us whether the relationship between advertising and sales is strong or weak, nor can it be meaningfully compared to the covariance between some other unrelated pair of variables measured in different units (say, temperature and ice cream sales).

Correlation solves exactly this problem. By dividing the covariance by the product of the two variables' standard deviations, correlation rescales the measure into a dimensionless quantity that always lies between -1 and $+1$, regardless of the original units. This makes correlation directly interpretable as a strength-of-relationship measure (close to ± 1 means a strong linear relationship, close to 0 means a weak one) and allows fair, apples-to-apples comparison of relationship strength across entirely different pairs of variables and measurement scales — something covariance's raw magnitude can never offer on its own.

3. Correlation vs Causation Trap

The Claim Being Made

A city reports that ice cream sales and drowning incidents have a correlation coefficient of $r = 0.85$ — a strong positive linear association — and a newspaper jumps from this number straight to the headline "ice cream causes drowning."

The Statistical Flaw

This is a textbook example of the correlation–causation fallacy. A correlation coefficient, however large, only quantifies the degree and direction of linear association between two variables measured together over the same time period or population — it is a purely descriptive, symmetric statistic that carries no information about mechanism, direction of influence, or cause-and-effect. Recall that Pearson's correlation is simply covariance rescaled by the two variables' standard deviations:

$$r = \text{Cov}(X,Y) / (SD_x \times SD_Y)$$

Nothing in this formula distinguishes "X causes Y" from "Y causes X" from "Z causes both X and Y" — r is completely symmetric in X and Y and blind to mechanism. $r = 0.85$ tells us that as one variable rises, the other tends to rise too, in a fairly consistent way; it does not, and cannot, tell us whether X causes Y, Y causes X, both are caused by some third factor, or the association is simply coincidental. Treating a high r -value as proof of causation ignores the very real possibility of a confounding (or "lurking") variable — something that independently drives both quantities up and down together, creating the appearance of a direct causal link where none exists.

The Likely Confounding Variable

Here, the obvious confounder is season / ambient temperature. During hot summer months, two independent things happen simultaneously: more people buy and eat ice cream because of the heat, and more people go

swimming, boating, and visiting beaches or pools to cool off — which mechanically increases the number of people exposed to drowning risk. Both ice cream sales and drowning incidents rise together in summer and fall together in winter, not because one causes the other, but because a third variable — temperature/season — is driving both. Eating ice cream does not impair swimming ability or otherwise physically cause drowning; the two series simply share a common seasonal rhythm.

What Additional Evidence Would Be Needed to Claim Causation

To move from "these two variables are correlated" to "one causes the other," much more evidence is required than a single correlation coefficient:

- Controlled experiment / randomization: ideally, randomly assign different levels of ice cream consumption to comparable groups and observe drowning outcomes while holding other factors (season, location, swimming exposure) constant. This is neither feasible nor ethical here, but it remains the conceptual gold standard for establishing causation.
- Temporal precedence: causation requires that the cause reliably occurs before the effect; the data would need to show ice cream consumption consistently preceding drowning incidents in a way that rules out simple simultaneity.
- Controlling for the confounder: statistically holding temperature/season constant — for instance using partial correlation or a regression model that includes season as a covariate — and checking whether the ice-cream–drowning association shrinks toward zero once season is accounted for. If it does, that confirms season, not ice cream, is doing the causal work.
- Dose–response relationship: evidence that higher levels of ice cream consumption are associated with proportionally higher drowning risk, in a consistent and graded way, would strengthen (though still not prove) a causal story.
- A plausible causal mechanism: some biological, physical, or behavioural pathway by which eating ice cream could actually lead to drowning — no such mechanism exists here, which itself is a strong argument against causation.
- Consistency across different contexts: a genuinely causal relationship should reproduce itself across different cities, countries, climates, and time periods, not just in this one dataset.

Without this kind of supporting evidence, $r = 0.85$ only establishes that the two variables move together — it does not establish why. The statistically sound conclusion is that ice cream sales and drowning incidents are both driven upward each summer by warmer weather and greater outdoor/water activity, and the newspaper's causal claim is unsupported by the correlation alone.

4. Effect of Outliers on Correlation

The Setup

A dataset of 9 points originally shows a Pearson correlation of $r = 0.42$, generally interpreted as only a weak positive linear relationship. After a single extreme outlier is identified and removed, the correlation recalculated on the remaining 8 points jumps sharply to $r = 0.89$ — a strong positive relationship. This dramatic swing, caused by removing just one data point out of nine, is itself the finding that needs to be interpreted.

What This Reveals About the Fragility of Pearson's Correlation

This result is a clear demonstration that Pearson's correlation coefficient is highly sensitive — fragile — to outliers, and that this fragility can badly mislead an analyst who trusts a single reported r -value without first inspecting the underlying scatterplot. Pearson's r is built from the sums of products of each point's deviation from the mean of its own variable; both the mean itself and every deviation used in the calculation are pulled around by extreme values. A single point that lies far from the rest of the cloud of data can dominate the numerator (and sometimes the denominator) of the correlation formula out of all proportion to the fact that it is just one observation among nine. In this case, that lone outlier was apparently working against the grain of the true underlying relationship — pulling the calculated r down from what would otherwise be a strong 0.89 relationship among the other eight points to a deceptively weak-looking 0.42 for the full dataset.

The broader lesson is that a single anomalous, possibly erroneous, data point can radically misrepresent not just the strength of a relationship but potentially its apparent direction as well (in more extreme cases, one outlier can even flip the sign of r). This is precisely why exploratory data analysis — plotting the data and visually screening for outliers — must always precede blind trust in a single summary correlation coefficient; reporting $r = 0.42$ without checking the underlying scatterplot would have led directly to the wrong substantive conclusion about how strongly these two variables are related.

Would Spearman's Rank Correlation Be More Robust Here? — Justification

Yes, Spearman's rank correlation would very likely be substantially more robust to this particular outlier, and the reasoning follows directly from how the two coefficients are constructed. Spearman's correlation does not operate on the raw numeric values of the variables at all; instead, it first converts each variable into ranks (1st smallest, 2nd smallest, and so on) and then applies Pearson's formula to those ranks. For n paired observations with rank-differences d , this simplifies to:

$$\rho = 1 - (6 \sum d^2) / (n(n^2 - 1))$$

An extreme outlier, no matter how far its raw value sits from the rest of the data, can still only occupy one rank position — the highest or lowest rank in the dataset — exactly as if it were merely a normal, unremarkable extreme value. Because ρ depends only on rank differences d , not on raw magnitudes, its influence on the rank-correlation calculation is capped by its position in the ordering, not amplified by its raw numeric distance from the other points the way it is under Pearson's formula.

Consequently, if this same 9-point dataset were re-analysed using Spearman's ρ , we would expect the coefficient to already be reasonably close to what it would be after the outlier's removal — the presence or

absence of the extreme point would not cause anywhere near the same dramatic 0.42-to-0.89 style of swing, because the relative ordering of the remaining eight points barely changes whether the ninth (extreme) point is included or not. In short, because Pearson's r is a moment-based, magnitude-sensitive statistic while Spearman's ρ is a rank-based, order-sensitive statistic, Spearman's correlation is the more robust and trustworthy choice whenever outliers are present or suspected in the data, and it is generally good practice to report both coefficients side by side when the presence of extreme values is a possibility.

5. Data Preparation — Missing Values

The Setup

A dataset of 200 rows has 15% missing values in the "Income" column (i.e., roughly 30 rows). Crucially, further analysis shows this missingness is not spread evenly across the dataset — it is more common among unemployed respondents. This means the pattern of missing data is systematic rather than random, and any handling strategy that ignores this pattern risks distorting the analysis.

(a) Classifying the Missing Data Mechanism

There are three standard categories of missing-data mechanism: MCAR (Missing Completely At Random), where missingness has no relationship to any variable, observed or unobserved; MAR (Missing At Random), where missingness is related to other observed variables in the dataset but not to the missing value itself, once those observed variables are accounted for; and MNAR (Missing Not At Random), where missingness depends on the value that is missing itself.

In this scenario, the missingness of Income is clearly not MCAR, since it is not occurring purely by chance — it is systematically concentrated among unemployed respondents. Because employment status is an observed variable in the dataset, and it appears to explain the pattern of missingness, this is best classified as MAR (Missing At Random): the probability that Income is missing depends on another variable we can actually see and measure (employment status), rather than on some completely unrelated or untraceable cause.

It is worth noting, however, that this case sits close to the boundary with MNAR. If the underlying reason unemployed respondents leave Income blank is tied to the value of their income itself — for instance, because they have zero or negligible income and feel there is nothing meaningful to report, or are embarrassed to disclose it — then the missingness mechanism would actually be driven by the unobserved income value, which is the defining feature of MNAR. In practice, since employment status is observed and can be explicitly conditioned on, analysts typically classify and treat this as MAR, but it is important to flag this ambiguity, because if a genuine MNAR component remains even after conditioning on employment status, no amount of clever imputation using only observed variables can fully remove the resulting bias.

(b) Why Mean Imputation Would Introduce Bias

Simple mean imputation works by replacing every missing Income value with a single number — the average income computed from the respondents who did report it:

$$\bar{X}_{\text{impute}} = \Sigma \text{Income}(\text{observed}) / n(\text{observed})$$

The problem is that the respondents who reported their income are disproportionately employed (since missingness skews toward the unemployed), so this $\bar{X}_{\text{impute}}$ is really an estimate of employed-respondent income, not of the population as a whole. Applying this same inflated figure to fill in the unemployed respondents' missing Income values would systematically overstate their true income, since unemployed individuals typically have lower — sometimes zero or near-zero — actual income.

This bias has several compounding effects on the analysis: it artificially inflates the recorded income of the unemployed subgroup, which understates true income inequality between employed and unemployed respondents in the dataset; it artificially shrinks the variance of the Income column, because a substantial block of values (15% of all rows) is replaced with one identical constant, making the data look far more homogeneous and tightly clustered than it truly is; and it weakens or distorts any statistical relationship between Income and other variables, such as employment status or education, since the imputed values do not reflect the real, lower income levels actually associated with unemployment. In short, mean imputation here would not just add noise — it would introduce a systematic, directional bias that misrepresents the very group the missingness is concentrated in.

(c) A Better Imputation Strategy

Because the missingness mechanism here is MAR and is explained by an observed variable — employment status — the correct remedy is to condition the imputation on that variable rather than imputing blindly from the overall column.

- Group-wise (conditional) imputation: impute each missing Income value using the mean or median of Income within its own employment-status subgroup, so unemployed respondents' missing incomes are filled in with values typical of other unemployed respondents, not the overall (employed-skewed) average.
- Regression / model-based imputation: build a regression model that predicts Income from other correlated variables available in the dataset — employment status, education, age, occupation, and so on — and use the model's predictions to fill in the missing values, capturing more of the true underlying structure than a simple group mean.
- Multiple Imputation (e.g., MICE — Multiple Imputation by Chained Equations): generate several plausible completed datasets by imputing each missing value multiple times with some random variation, run the intended analysis on each, and pool the results. This is the most statistically rigorous approach for MAR data because it also preserves and reflects the genuine uncertainty introduced by the missing values, rather than pretending the imputed numbers are known with certainty.
- Add a "missingness" indicator variable: alongside whichever imputation method is chosen, include a binary flag recording whether each row's Income was originally missing. This allows any downstream

model to separately learn from the fact of missingness itself, which helps guard against any residual MNAR-like effect that conditioning on employment status alone might not fully capture.

Given that missingness here is tied specifically to employment status, at an absolute minimum the imputation must condition on employment status rather than use a single overall mean; a regression-based or multiple-imputation approach that explicitly uses employment status (and ideally other correlated variables) as predictors would be the most defensible and accurate strategy, since it directly addresses the mechanism causing the data to be missing in the first place.